

A Course in Analytic Number Theory

Marius Overholt

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in Mathematics**

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To the memory of my father Finn and mother Liv

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Preface

This book was written for graduate students looking for an introduction to some basic methods of analytic number theory. It is suitable as a textbook for an introductory one-semester course at the beginning graduate level, but contains more material than can be comfortably covered in such a course. However, by suitably selecting chapters, it is possible to teach courses going in various directions.

Readers should be familiar with ε - δ calculus, have completed an undergraduate course in complex analysis, and possess the proficiency in abstract and linear algebra to be expected of a beginning graduate student. No familiarity with graduate-level analysis is assumed. The first four chapters presuppose no complex analysis beyond simple properties of the exponential function.

Each chapter is followed by notes with references and historical remarks. At the end of many of the sections there are references to more detailed treatments of the topic under consideration.

I wish to thank anonymous referees for suggestions that have improved the book. Naturally I alone remain responsible for all errors and imperfections that remain.

This seems an appropriate place to express my gratitude to Hugh Montgomery, Imre Ruzsa, and the late Sigmund Selberg, from whose teaching of analytic number theory I have benefited.

Marius Overholt

Acknowledgments

Except for some exercises, I am indebted to the literature of analytic number theory for all the material in this textbook.

As references I have chiefly relied on the following works: For arithmetic functions *Introduction to Analytic and Probabilistic Number Theory* by Gérald Tenenbaum, and for prime number theory *Multiplicative Number Theory I. Classical Theory* by Hugh L. Montgomery and Robert C. Vaughan have been the main references. But for an easy proof of the Prime Number Theorem with an error term I have followed *The Distribution of Prime Numbers* [Ing90] by A. E. Ingham. For my very modest account of the analytic properties of the Riemann zeta function I am indebted to *The Theory of the Riemann Zeta-function* [Tit86] by E. C. Titchmarsh. The chapter on the Circle Method owes the most to the second edition of *Analytic Methods for Diophantine Equations and Diophantine Inequalities* [Dav05] by H. Davenport, edited by T. D. Browning and with a Foreword by D. E. Freeman, D. R. Heath-Brown and R. C. Vaughan, though in a few particulars I have followed the treatment in the second edition of *The Hardy-Littlewood Method* [Vau97] by R. C. Vaughan. For the chapter on the Dedekind zeta function my main sources have been *Lectures on Algebraic and Analytic Number Theory* [Gál61] by I. S. Gál, the contribution by H. A. Heilbronn in the collection *Algebraic Number Theory* [Hei67] edited by J. W. S. Cassels and A. Frölich, *Algebraic Number Theory* [Lan70] by Serge Lang, *Elementary and Analytic Theory of Algebraic Numbers* [Nar00b] by W. Narkiewicz, *Algebraic Number Theory* by Jürgen Neukirch, and especially the expository articles *The Analytic Theory of Algebraic Numbers* [Sta75] and *Galois Theory, Algebraic Number Theory and Zeta Functions* [Sta95] by H. M. Stark.

Beyond these works that I have mainly relied on for this book, there are many other excellent treatments of analytic number theory. It may be appropriate at this point to mention a few that are particularly important for one reason or another. *Introduction to Analytic Number Theory* by Tom M. Apostol [**Apo76**] has for decades been the most widely used introductory text. It carefully develops material needed from elementary number theory rather than assuming it as a prerequisite, and has many exercises. *Multiplicative Number Theory* by Harold Davenport [**Dav00**] is a classic account of the distribution of primes in arithmetic progressions. It has been in print for more than forty years, and is still one of the more frequently assigned textbooks for courses in analytic number theory. *Analytic Number Theory* by Henryk Iwaniec [**IK04**] and Emmanuel Kowalski is a broad, deep and modern treatment. Of outstanding importance to the development of analytic number theory in its early stages was *Handbuch Der Lehre von der Verteilung der Primzahlen* [**Lan74**] by Edmund Landau.

For information on the historical background I have in addition relied on these sources: *Gauss and Jacobi Sums* [**BEW98**] by B. C. Berndt, R. J. Evans and K. S. Williams, *Pioneers of Representation Theory: Frobenius, Burnside, Schur, and Brauer* [**Cur99**] by C. W. Curtis, the survey paper [**DE80**] by H. G. Diamond on elementary methods in prime number theory, *History of the Theory of Numbers* [**Dic34**] by L. E. Dickson, *Riemann's Zeta Function* [**Edw01**] by H. M. Edwards, the sixth edition of *An Introduction to the Theory of Numbers* [**HW08**] by G. H. Hardy and E. M. Wright and revised by D. R. Heath-Brown and J. H. Silverman, the introductory notes by H. A. Heilbronn in volume I of the Collected Papers of G. H. Hardy [**Har66**], the 1990 edition of *The Distribution of Prime Numbers* [**Ing90**] by A. E. Ingham and with a Foreword by R. C. Vaughan, *Multiplicative Number Theory I. Classical Theory* [**MV07**] by H. L. Montgomery and R. C. Vaughan, and *The Development of Prime Number Theory* [**Nar00a**] by W. Narkiewicz, and the survey paper [**VW02**] on Waring's Problem by R. C. Vaughan and T. D. Wooley.

All figures have been made with InkscapeTM and MathematicaTM.

Marius Overholt

How to use this text

Chapters 1, 2, 3, 5 and 6 are suitable for a course emphasizing arithmetic functions and the two classical highlights of analytic number theory: Dirichlet's theorem on primes in arithmetic progressions, and the Prime Number Theorem.

Chapters 1, 2, 3 and 4 are suitable for a syllabus with an emphasis on elementary methods, for students with little knowledge of analysis. Unfortunately the Prime Number Theorem is not covered.

Chapters 1, 3, 4, 5 and 6 are suitable for a syllabus with a Diophantine emphasis, but also including a proof of the Prime Number Theorem.

Chapters 1, 3, 5, 6, 8 and 10 are suitable for a syllabus with an emphasis on analytic methods, concentrating on the Riemann zeta function and the distribution of primes.

Chapters 1, 3, 5, 8 and 9 are suitable for a syllabus with an emphasis on the analytic theory of number fields. Students following such a syllabus should either have some knowledge of algebraic number theory, or else have a good knowledge of abstract algebra and do some reading. Here the Prime Number Theorem is established by means of the Ikehara theorem.

Chapters 1, 3, 5 and 7 are suitable for a syllabus aiming at the Prime Number Theorem for arithmetic progressions. The ordinary Prime Number Theorem is established as a corollary at the end of the course.

To help with the planning of syllabi a diagram of dependencies between chapters has been provided; see Figure 1 on page xvi. To supplement this, more specific comments may prove useful: Unless the material on Artin L-functions in Chapter 9 is covered, Section 3.8 may be substituted for Sections 3.4 and 3.5. This saves a little time, and the construction of Dirichlet

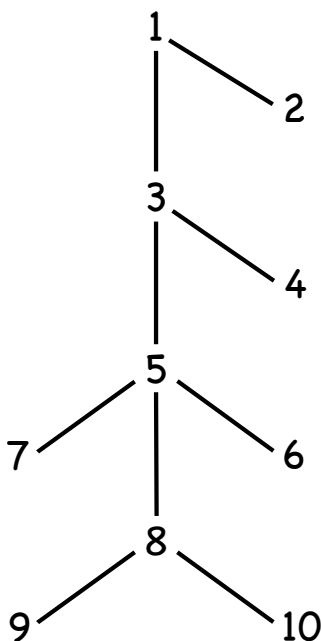


Figure 1. Diagram of chapter dependencies

characters by means of primitive roots seems less conceptually demanding than even an epsilon of representation theory. The theorem of Siegel from Section 7.3 is also needed in Section 9.3. The Borel-Carathéodory lemma of Section 7.4 is also used to prove the Hadamard factorization theorem of Section 8.6. The Hadamard factorization theorem is needed in Section 9.4 as well as in Chapter 10. As usual, starred material may be omitted without loss of continuity.

There are exercises at the end of the chapters, for which solutions are not provided. In addition to the end-of-chapter exercises, there is a chapter with a selection of other exercises with solutions. As a caution, some exercises have been marked with a dagger, because they require more work than the remainder. But they are for the most part not more difficult in the sense of it being harder to see what to do.

A *Summary of Elementary and Algebraic Number Theory* with a condensed exposition of those concepts on which the book draws is available on the web. The Summary presupposes familiarity with groups, rings and fields. All results from elementary and algebraic number theory that are actually needed in this book are proved in the Summary.

Introduction

Analytic number theory is mainly devoted to finding approximate counts of number theoretical objects in situations where exact counts are out of reach. Primes, divisors, solutions of Diophantine equations, lattice points within contours, partitions of integers and ideal classes of algebraic number fields are some of the objects that have been counted. The prototypical approximate count in number theory is the Prime Number Theorem (PNT), stating that

$$\lim_{x \rightarrow +\infty} \frac{\pi(x)}{\int_2^x \frac{du}{\log(u)}} = 1$$

where $\pi(x)$ is the number of primes $p \leq x$. This was proved independently in 1896 by Jacques Hadamard and Charles de la Vallée Poussin, building on ideas of Bernhard Riemann, and applying complex analysis to the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

to establish the result. An asymptotic count like the PNT usually attracts attention with a view to improve it. As the distribution of prime numbers is one of the central topics in number theory, much effort has been expended to obtain improvements to the Prime Number Theorem. We shall prove one of the weaker ones, to the effect that there exist positive constants c, C, x_0 such that

$$\left| \pi(x) - \int_2^x \frac{du}{\log(u)} \right| \leq Cx e^{-c \log^{1/10}(x)} \quad \text{for } x \geq x_0.$$

This is more precise, though also more complicated to state, than the asymptotic form of the PNT.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$d(n)$	1	2	2	3	2	4	2	4	3	4	2	6	2	4	4

Table 1. Values of the divisor function

Counting the number of divisors of positive integers leads to a difficult problem known as the Dirichlet Divisor Problem that is still unsolved today. Denoting the number of divisors of n by $d(n)$, Table 1 shows that these counts fluctuate a good deal. But much more regular behavior is revealed by averaging $d(n)$, and in fact

$$\frac{1}{x} \sum_{n \leq x} d(n) \approx \log(x) + 2\gamma - 1, \quad \gamma = 0.5772 \dots,$$

with an absolute error that tends to zero as $x \rightarrow +\infty$. To determine how fast the error tends to zero is the divisor problem of Dirichlet.

Divisors and primes are the stuff of multiplicative number theory. But there are also interesting counting problems connected with additive questions. The eighteenth-century English algebraist Edward Waring stated that every positive integer may be expressed as a sum of a limited number of k -th powers of nonnegative integers, the number required depending on k only. We shall count the number of such representations for large integers when the number of powers allowed is sufficiently large, finding an asymptotic formula by means of the Circle Method and establishing Waring's claim. This was first achieved by David Hilbert by a method different from the one used here. The proof is the most elaborate in the book, though the prerequisites are surprisingly modest. The Circle Method is related to Fourier theory, but involves only Fourier series with finitely many terms so convergence issues do not arise.

The achievements of analytic number theory are not entirely limited to approximate counts. Some of the quantities estimated are not counting numbers, and for a few problems exact rather than approximate results have been attained. We shall cover one such case from algebraic number theory, that of the analytic class number formula

$$h_K = \frac{w_K |d_K|^{1/2}}{2^{r_1(K)+r_2(K)} \pi^{r_2(K)} R_K} \lim_{s \rightarrow 1} \frac{\zeta_K(s)}{\zeta(s)}$$

that expresses the number h_K of ideal classes of the ring of algebraic integers of a number field K in terms of other arithmetic data. This formula is due to Dirichlet and Richard Dedekind.

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List of Notations

A	limit superior of $\psi(x)/x$, page 4
a	limit inferior of $\psi(x)/x$, page 4
a	nonzero ideal, page 255
$A(x)$	counting function $A(x) = \sum_{A \ni n \leq x} 1$ of a sequence A , page 43
$a_K(m)$	the number of ideals in $\mathcal{O}_K$ whose norm equals m , page 255
$A_n(q)$	an exponential sum in the Circle Method, page 120
$\mathfrak{b}'$	the Dedekind complement of $\mathfrak{b}$, page 260
$\mathbb{C}$	the complex plane, page 3
$\mathcal{C}l_K$	the class group of $\mathcal{O}_K$, page 257
$C_{k,s}$	a constant in the Circle Method, page 121
$c_q(n)$	Ramanujan sum, page 89
$\deg(\rho)$	degree of a group representation ρ , page 71
$\mathbb{D}$	unit disk, page 206
dA	asymptotic density of a sequence A , page 43
D	discriminant $D = b^2 - 4ac$ of a binary quadratic form, page 100
$d(n)$	divisor function, page 16
$D(x)$	summatory function $D(x) = \sum_{n \leq x} d(n)$, page 23
d_K	the discriminant of a number field K , page 257
$d_k(n)$	number of ways of writing n as a product of k factors, page 103
$e(n)$	neutral element under Dirichlet convolution, page 17
$e_k(n)$	indicator function of $\{k\}$, page 20
$e(x)$	exponential function $e(x) = \exp(2\pi ix)$ of period 1, page 68

$E_J(z)$	Weierstrass primary factor, page 235
$\hat{f}$	Fourier transform of f , page 213
$\hat{f}_n$	n -th Fourier coefficient of f , page 210
$\text{Frob}_{\mathfrak{P}}$	Frobenius element, page 296
$\mathfrak{f}_\chi$	conductor of the Dirichlet character χ , page 91
$f(\alpha)$	an exponential sum in the Circle Method, page 117
$f_j(\alpha)$	an exponential sum in the Circle Method, page 117
F_n	Fermat number, page 108
$\hat{G}$	dual of a finite group G , page 79
$\text{GL}(V)$	general linear group, page 71
$G(k)$	least s in Waring's problem for k -th powers for large n , page 112
$g(k)$	least s in Waring's problem for k -th powers for all n , page 112
$\mathbb{H}$	upper half plane of points τ with $\text{Im}(\tau) > 0$, page 216
$\text{Ind}_H^G(\rho)$	the representation induced on G by ρ on a subgroup H , page 293
$\mathbf{I}_A(x)$	indicator function of a sequence A , page 43
$\mathbf{I}_{q,a}(n)$	indicator function of the arithmetic progression $a + q\mathbb{Z}$, page 84
$\text{id}(n)$	identity function $\text{id}(n) = n$, page 16
J_K	the group of nonzero fractional ideals of $\mathcal{O}_K$, page 257
$J_\rho(n)$	an integral in the Circle Method, page 120
K	an algebraic number field, page 255
$\liminf$	limit inferior, page 4
$\limsup$	limit superior, page 4
$\text{Li}(x)$	integral logarithm $\text{Li}(x) = \text{PV} \int_{[0,x]} du/\log(u)$, page 246
$L(s, \chi)$	Dirichlet L-function $L(s, \chi) = \sum_{n \in \mathbb{N}} \chi(n)n^{-s}$, page 85
$\text{li}(x)$	integral logarithm $\text{li}(x) = \int_{[2,x]} du/\log(u)$, page 1
$(\mathcal{M}f)(s)$	Mellin transform of f , page 227
$\mathfrak{M}$	the major arcs intervals in the Circle Method, page 117
$\mathfrak{m}$	the minor arcs intervals in the Circle Method, page 117
$\mathfrak{M}(q, a)$	a single major arcs interval in the Circle Method, page 117
$M(x)$	summatory function $M(x) = \sum_{n \leq x} \mu(n)$, page 50
$M_n(q)$	number of solutions of a congruence for a diagonal form, page 113
$N(\alpha)$	norm of an algebraic number α , page 255
$N(\mathfrak{a})$	norm of an ideal $\mathfrak{a}$, page 255
$N(\sigma, T)$	number of zeros ρ of $\zeta(s)$ with $\beta > \sigma$ and $0 < \gamma \leq T$, page 323

$n(p)$	least positive quadratic nonresidue modulo p , page 94
$N(T)$	number of zeros of $\zeta(s)$ with $0 \leq \sigma \leq 1$ and $0 < t \leq T$, page 309
$N_0(T)$	number of zeros $s = 1/2 + it$ of $\zeta(s)$ with $0 < t \leq T$, page 245
n_K	the degree of K over $\mathbb{Q}$, page 255
$\mathcal{O}_K$	ring of algebraic integers in a number field K , page 255
$O(g)$	big-O notation of Bachmann, page 7
$o(g)$	small-o notation of Landau, page 7
$\mathfrak{p}$	nonzero prime ideal, page 255
$p(n)$	partition function, page 104
$P^+(n)$	the largest prime factor of n , page 37
$P^-(n)$	the smallest prime factor of n , page 37
P_K	the subgroup of J_K of principal fractional ideals, page 257
$\mathbb{Q}$	the field of rational numbers, page 149
$Q(x)$	counting function $Q(n) = \sum_{n \leq x} \mu(n) $ of the squarefrees, page 42
$Q(x, y)$	binary quadratic form $Q(x, y) = ax^2 + bxy + cy^2$, page 100
$\mathbb{R}$	the field of real numbers, page 68
$\text{Res}_H^G(\rho)$	the restriction of ρ from G to a subgroup H , page 295
r	$= r_1 + r_2 - 1$ is the rank of the group of units of $\mathcal{O}_K$, page 264
$R(n)$	number of solutions in the Circle Method, page 116
r_1	the number of real embeddings of a number field, page 257
r_2	half the number of complex embeddings, page 257
R_K	the regulator of $\mathcal{O}_K$, page 267
$\text{rad}(n)$	radical $\text{rad}(n) = \prod_{p n} p$ of n , page 16
$\mathfrak{S}(n)$	the singular series in the Circle Method, page 123
S	Weyl sum, page 130
s	notation $s = \sigma + it$ for a complex variable, page 62
$S(q, a)$	an exponential sum in the Circle Method, page 114
$S(x)$	sawtooth function $S(x) = x - [x] - 1/2$, page 11
t	imaginary part of the complex variable $s = \sigma + it$, page 62
$T(x)$	summatory function $T(x) = \sum_{n \leq x} \log(n)$, page 3
$T_{\mathfrak{p}}$	inertia group, page 296
$\mathfrak{U}$	an interval of integration in the Circle Method, page 117
U_K	the group of units of $\mathcal{O}_K$, page 264
$v(\beta)$	an integral in the Circle Method, page 117

$v_j(\beta)$	an integral in the Circle Method, page 120
V_ρ	representation space of a group representation ρ , page 71
$W(\chi)$	root number in the functional equation of $L(s, \chi)$, page 231
w_K	the number of units of finite order in $\mathcal{O}_K$, page 264
$Z(s)$	completed zeta function $Z(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, page 230
$Z_K(s)$	the completed Dedekind zeta function, page 257
$Z_K(s, C)$	the completed zeta function of an ideal class C , page 257
$Z_{\mathfrak{P}}$	decomposition group, page 296
β	real part of a zero ρ of $\zeta(s)$ in $\sigma > 0$, page 170
χ	Dirichlet character, page 84
χ_0	principal Dirichlet character, page 85
χ_0	trivial character, page 76
χ_ρ	character of a group representation ρ , page 76
δA	logarithmic density of a sequence A , page 44
$\Delta(x)$	error term in $D(x) = x \log(x) + (2\gamma - 1)x + \Delta(x)$, page 24
δ_{il}	Kronecker delta $\delta_{il} = 1$ if $i = l$, zero otherwise, page 75
γ	Euler-Mascheroni constant, page 12
γ	imaginary part of a zero ρ of $\zeta(s)$ in $\sigma > 0$, page 170
$\Gamma(s)$	gamma function, page 223
$\gamma_K(s)$	the gamma factor for the number field K , page 257
λ	rank of an entire function, page 238
$\Lambda(n)$	von Mangoldt function, page 2
$\lambda(n)$	Liouville function $\lambda(n) = \prod_{p^k n} (-1)$, page 16
μ	exponent of convergence for an entire function, page 237
$\mu(n)$	Möbius function, page 18
$\Omega(n)$	total number $\Omega(n) = \sum_{p^k n} 1$ of prime factors of n , page 16
$\omega(n)$	number of distinct prime factors $\omega(n) = \sum_{p n} 1$ of n , page 16
$\Phi(\lambda, \alpha, s)$	Lerch zeta function, page 251
$\Phi(x)$	summatory function $\Phi(x) = \sum_{n \leq x} \phi(n)$, page 21
$\phi_2(q)$	number of primitive characters modulo q , page 92
$\Pi(x)$	weighted counting function $\Pi(x) = \sum_{p^k \leq x} k^{-1}$, page 246
$\pi(x)$	counting function $\pi(x) = \sum_{p \leq x} 1$ of the primes, page 1
$\Psi(s; q, a)$	a Dirichlet series, page 183
$\psi(x)$	Chebyshev function $\psi(x) = \sum_{p^k \leq x} \log(p)$, page 2

$\psi(x; q, a)$	$= \sum_{x \geq n \equiv a \pmod q} \Lambda(n)$, page 183
$\psi_1(x)$	smoothed Chebyshev function, page 173
ρ	group representation, page 71
ρ	zero of $\zeta(s)$ in $\sigma > 0$, page 170
ρ_0	trivial group representation, page 71
σ	real part of the complex variable $s = \sigma + it$, page 62
$\sigma(n)$	sum-of-divisors function $\sigma(n) = \sum_{d n} d$, page 17
σ_a	abscissa of absolute convergence of a Dirichlet series, page 66
$\sigma_a(A)$	abscissa of absolute convergence of the Dirichlet series of A , page 67
σ_c	abscissa of convergence of a Dirichlet series, page 65
$\sigma_c(A)$	abscissa of convergence of the Dirichlet series of A , page 67
$\tau(\chi, n)$	Gauss sum of a Dirichlet character χ , page 89
τ_p	classical Gauss sum, page 70
ϱ	order of an entire function, page 237
$\vartheta(x)$	weighted counting function $\vartheta(x) = \sum_{p \leq x} \log(p)$ of the primes, page 2
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$\vartheta_{jk}(z, \tau)$	Jacobi theta function, page 249
$\tilde{\rho}$	the inflation of ρ from G/H to G , page 296
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$\Xi(z)$	Riemann Xi function $\Xi(z) = \xi(1/2 + iz)$, page 230
$\zeta(s)$	Riemann zeta function $\zeta(s) = \sum_{n \in \mathbb{N}} n^{-s}$, page 63
$\zeta_K(s)$	Dedekind zeta function of a number field K , page 256
$\zeta_K(s, C)$	the zeta function of an ideal class C , page 257
$\delta_{A,B}$	Dirichlet density of A in B , page 88
$1(n)$	constant arithmetic function $1(n) \equiv 1$, page 17
$[x]$	integer part function $[x] = \max\{n \in \mathbb{Z} \mid n \leq x\}$, page 2
$*$	Dirichlet convolution $(f * g)(n) = \sum_{d n} f(d)g(n/d)$, page 17
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$\ll$	Vinogradov notation, page 7
$\overline{n}$	multiplicative inverse modulo q , page 70
$\sim$	asymptotic equality $f(x) \sim g(x)$, page 1

$\varphi \mapsto \check{\varphi}$	inverse Fourier transform operator, page 79
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